Data-driven recommendations for increasing reliability in measures of infant word recognition

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Abstract

Children's early word recognition is both an important phenomenon in its own right and a key index of early language more broadly. A central paradigm for measuring word recognition in young children is the looking-while-listening paradigm (LWL), which tracks eye movements to the referent of a word as a measure of online language comprehension. Yet the analytic and design choices for LWL vary widely, and the consequences of these choices have rarely been investigated at a large scale. Here we use data from Peekbank, a repository of LWL data from studies with almost 4000 children (6 months–5 years), to search for analytic and design decisions that maximize the reliability and validity of LWL measures. Analytic recommendations include using long analytic windows for computing accuracy and avoiding baseline correction. We also find little or no benefit from excluding children or trials with low data. The recommended combination of analytic choices increased reliability (intra-class correlations and test-retest correlations) by approximately .20 and validity (correlations with productive vocabulary) by .07. Effects were generally consistent across age, but gains for reliability were more substantial than for validity. Design recommendations include building experiments with a minimum of 10 target trials per child per condition and matching stimuli for animacy. We hope that these recommendations lead to improved and unified standards for the measurement of children's word recognition in the LWL paradigm.

Keywords: looking while listening, word recognition, language development, eyetracking, reliability, validity

Word count: X

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Introduction

Over the first few months of life, children become attuned to their native language, organizing the phonological system of speech input and parsing speech into meaningful units. Soon after, children begin to show early signs of comprehension of words or phrases through simple behaviors such reaching for an object that is named (e.g., "Where's the ball?"). These everyday accomplishments mark important milestones in children's development. Yet global assessments of these milestones do not capture the millisecond-level processes that take place during real-time language comprehension. Children, like adults, process speech incrementally, as it unfolds in time (Fernald, Pinto, Swingley, Weinberg, & McRoberts, 1998; Frank et al., 2026; Snedeker, 2013), and use features of the speech signal to extract meaning that supports language comprehension (Law & Edwards, 2015; Weisleder & Fernald, 2013). One powerful, child-friendly method for tracking these moment-to-moment processes is precisely measuring looking behavior in response to speech. In particular, the "looking while listening" (LWL) procedure has emerged as a *de facto* standard for assessing language comprehension in infants and children (Frank, Braginsky, Yurovsky, & Marchman, 2021).[1] In the standard version of the LWL task, children see two pictures on a screen and hear speech directing their attention to one of the pictures, e.g., "look at the ball". The accuracy and speed of fixating the named referent measure the efficiency of language comprehension (Fernald et al., 2008).

The LWL procedure has been a key tool for enhancing our understanding of the continuous processes of growth in incremental language processing (Golinkoff, Ma, Song, & Hirsh-Pasek, 2013). For example, using LWL methods, we can measure developmental increases in the speed and accuracy of children's recognition of familiar words (Fernald, Perfors, & Marchman, 2006; Frank et al., 2026) and in their sensitivity to detailed phonological information during the first years after birth (Moore & Bergelson, 2024; Swingley & Aslin, 2000). Moreover, measures of children's skill in the LWL task have proven useful as an index of individual differences. Children who show more efficient language processing of familiar words at 18 months show larger vocabulary size and more rapid vocabulary growth over the second and third years of life than children with less efficient processing (Fernald & Marchman, 2012; Fernald et al., 2006; Frank et al., 2026; Peter et al., 2019). Variation in language processing efficiency has also been shown to be associated with later language, cognitive, and non-verbal outcomes in both typically-developing (Law & Edwards, 2015; Peter et al., 2019) and preterm children (Loi, Marchman, Fernald, & Feldman, 2017; Marchman, Adams, Loi, Fernald, & Feldman, 2016; Marchman et al., 2018).

Despite widespread use of the LWL method in language development research, there is still limited consensus on how to best process and analyze time-series data of this type (from either adults or infants, Mirman, Dixon, & Magnuson, 2008; Bergelson & Swingley,

[1] Here we adopt LWL as a convenient label while recognizing that the procedure goes by a number of different names in the literature, including the "intermodal preferential looking procedure" (Fernald, Zangl, Portillo, & Marchman, 2008; Hirsh-Pasek & Golinkoff, 1996; Reznick, 1990).

2012; Peelle & Van Engen, 2021; Von Holzen & Mani, 2012; Weaver, Saffran, & Arunachalam, 2026). This lack of consensus likely stems from structural factors: infant recruitment is time- and cost-intensive and datasets collected by individual labs are often small (Bergmann et al., 2018; Frank, Braginsky, Yurovsky, & Marchman, 2017). These issues make it difficult for any one researcher to systematically evaluate the consequences of specific analytic choices independently of the data collected for a particular study (cf. Peelle & Van Engen, 2021). Instead, researchers make ad-hoc decisions and adopt idiosyncratic lab practices that themselves originate from subjective judgments of what "seems to work". By decreasing uniformity in the literature, this variation also increases the challenge of triangulating what precisely is being measured by the LWL task (Weaver et al., 2026). How can we put the measurement and analysis of infant word recognition and online language processing on firmer empirical footing?

Our approach

Peekbank is a new data resource that offers a potential solution to this core issue by providing a large database of infant eye-tracking data from the LWL paradigm for exploring methodological questions (Zettersten et al., 2023). The most recent release of Peekbank (version 2026.1) houses 44 datasets from almost 4000 infants across a wide age range (6 months–5 years) and across a broad set of test words and materials. The goal of the current manuscript is to use these data to systematically explore key analytic decisions that researchers make when using this paradigm, offering best-practices guidance for measurement.

Our approach throughout is to seek analytic choices that distill the best possible measures from the rich time course of behavior in the LWL task. Recent reviews of the LWL literature highlight open questions about both the reliability and validity of looking-based measures of word recognition (Weaver et al., 2026). Starting with reliability, we aim for analyses that yield the maximal signal about individual children relative to both measurement error and irrelevant artifacts (Byers-Heinlein, Bergmann, & Savalei, 2022; Rhemtulla & Bottesini, 2022; Schreiner et al., 2024). Here, to measure reliability, we compute both within-dataset inter-item reliability and test-retest reliability for datasets with multiple LWL sessions close in time for the same children.

Turning to validity, the goal is to make analytic decisions that maximize the relationship with measures of word knowledge (Flake & Fried, 2020; Kominsky, Lucca, Thomas, Frank, & Hamlin, 2022). Here, we compute the correlations between LWL measures and vocabulary using aggregate measures of receptive and expressive vocabulary size from the MacArthur-Bates Communicative Development Inventories (MB-CDI). The MB-CDI is a popular parent report measure of early vocabulary (Frank et al., 2021; Marchman, Dale, & Fenson, 2023) that is widely used in conjunction with LWL tasks (Fernald & Marchman, 2012; Fernald et al., 2006; Marchman & Fernald, 2008). Although ideally we would have access to measures of precisely the same construct (e.g., button-press reaction times for word recognition), in practice, such data are not available at the scale we require and we thus rely on the CDI as a frequently-used measure of related but distinct constructs.

Our focus in this paper is to provide guidance for navigating some of the many decisions facing an investigator analyzing LWL data. Previous work has provided in-depth recommendations for experiment design, eye-tracking, and eye-gaze coding (Fernald et al., 2008; Olson et al., 2020; Venker et al., 2020; Wass, Smith, & Johnson, 2013). Other recent work using an earlier version of Peekbank (as well as other datasets) has also investigated the psychometric properties of a range of different dependent measures in LWL tasks and their general construct structure (Sander-Montant, Fibla, Killam, & Byers-Heinlein, 2026). We focus here instead on the consequences of varying analytic choices for two main measures which have the strongest psychometric properties (Sander-Montant et al., 2026). The first is *accuracy*: proportion looking to the target image (out of all looks to either the target or distractor image) computed over some period of time after the onset of the target word. The second measure is shift latency or *reaction time* (RT): how long it takes a child to shift their gaze from the distractor to the target image in response to the target word (Fernald et al., 2008).

Considering these measures, we address the following data processing (1-4) and design (5-6) questions with the goal of formulating broad recommendations. For each question, we also examine whether the results are modulated by age.

1. For accuracy, what start and end time maximize the signal of children's response to the stimulus?
2. For accuracy, should analysts correct for baseline preferences in looking to the target image prior to the onset of the target word?
3. When computing RTs, it best to operationalize RT as the latency from target word onset to when the child initiates an eye movement away from the distractor picture (launch RT) or to when a child completes this shift and fixates the target picture (landing RT)? What analytic window should be used? And should RTs be subject to a logarithmic transformation?
4. For both accuracy and RT, when should participants and trials be excluded from the analysis?
5. How many trials are needed to maximize reliability?
6. How do design choices in pairing target and distractor items affect measurement?

To foreshadow our results, we find that while many common data analysis practices provide reasonable levels of reliability and validity, it may be possible to further optimize reliability and validity through careful, data-driven decision-making. Overall, we find data-driven analytic choices that lead to gains in both reliability (~.20) and validity (.07); we return in the discussion to why gains in reliability might be more substantial.

Methods: Dataset and approach

Dataset

Peekbank is an aggregation of datasets from a number of LWL studies. This work is based on Peekbank version 2026.1 (released February 2026) which houses 44 datasets containing 134,929 trials from 3,915 children participating in 5,596 distinct testing sessions (some children participated in multiple sessions).

We limit our analyses to what we term "standard" trials, resulting in 75,636 trials from 3,171 children participating in 4,851 sessions across 36 datasets. For a trial to be considered "standard", both the target and the distractor images must be of familiar objects. For the auditory stimulus, the target word must be a real word that occurs once and is the first and only source of information about the target. In our analyses, the target word is also always a noun. The carrier phrase must be a well-formed, grammatical phrase in a single language (e.g., English, Spanish), without any intentionally-added noise or audio filtering. We focus on these trials because they are commonly used and most directly reflect the construct of incremental word recognition processes. (See SI: Criteria for Inclusion for more detailed criteria used to handle borderline situations.)

Following Peekbank's convention, trial time is centered at the onset of the target word (labeled as 0 ms). We only include trials from children 60 months or younger; the average age in the final dataset was 22 months (SD: 9 months, median: 19 months; see Figure S1 for age histograms). The final dataset included 1,856 girls, 1,927 boys, and 132 children of other or unspecified gender. Datasets were primarily collected in the United States, but also in Canada, Mexico, the Netherlands, and Norway. No additional demographic information is available in the Peekbank database; more detailed demographic information on each dataset can be found in the corresponding papers.

MB-CDI comprehension scores were available for 632 children from 12 datasets, and MB-CDI production scores were available for 1,143 children from 20 datasets. Test-retest data comes from 5 datasets with multiple sessions within a month of each other, covering 365 children (788 session pairs). See Table 1 for a list of included datasets. Figure S2 has time course plots showing the number of trials for each dataset.

Approach

Reliability refers to how consistently measurements of the same thing produce the same outcome. One measure of reliability is an intra-class correlation (ICC) that measures (within a dataset) how much different items (target labels; "raters" in ICC terminology) are consistent with each other in measuring a child within a session. Specifically, we computed ICCs based on a mean-rating, consistency, two-way random effects model (Type 2A in Gwet (2014)'s typology) using the agreement package in R (Girard, 2019) (see SI: Details about ICC for more specifics). Another measure of reliability is test-retest reliability, which measures how similarly the same child performs when assessed on two different occasions on the same task. This measure is only possible on a subset of the datasets that repeated sessions at closely spaced intervals. While test-retest reliability data are limited, this is the metric that is most important for measuring the stability of individual differences.

Validity refers to whether measures correspond to the construct of interest (in this case, language processing efficiency and/or receptive/expressive vocabulary size). We focus specifically on the relationships between aggregate measures of receptive and expressive vocabulary size from the MB-CDI and LWL task, given the documented relationships between these measures (e.g., Fernald & Marchman, 2012; Peter et al., 2019). This correspondence reflects the general relationship between familiar word recognition in the

Table 1

Descriptives of the included datasets. Age is mean age in months, with the range shown in brackets.

	Dataset Name	Pct Trials	Trials	Sessions	Subjects	Age	CDI	Test-retest	Language
1	Adams et al. (2018)	18.4%	13,899	711	69	22 [13 - 38]	x	x	English
2	Fernald & Marchman (2012)	15.6%	11,828	679	122	22 [17 - 32]	x	x	English
3	Weaver et al. (2024)	6.5%	4,948	248	141	16 [14 - 24]		x	English
4	Fernald et al. (2013)	5.8%	4,359	178	80	20 [17 - 26]	x	x	English
5	Hurtado et al. (2008)	5.3%	4,035	136	76	21 [17 - 27]	x		Spanish
6	Fernald et al. (2006)	4.9%	3,692	229	63	20 [15 - 25]	x		English
7	Yurovsky et al. (2013)	3.7%	2,810	413	413	34 [12 - 60]			English
8	Yurovsky et al. (2017)	3.1%	2,350	327	327	37 [8 - 60]			English
9	Bergelson & Swingley (2012)	2.8%	2,081	84	84	12 [6 - 21]	x		English
10	Kartushina & Mayor (2019)	2.7%	2,072	69	69	7 [6 - 9]	x		Norwegian
11	Weaver & Saffran (2026)	2.4%	1,788	69	69	19 [18 - 20]	x		English
12	Weisleder & Fernald (2013)	2.3%	1,769	58	29	22 [18 - 27]			Spanish
13	Yurovsky & Frank (2017)	2.3%	1,747	285	285	26 [13 - 59]			English
14	Borovsky & Peters (2019)	2.2%	1,676	79	79	18 [17 - 20]	x		English
15	Potter & Lew-Williams (2023)	2.1%	1,626	67	67	24 [21 - 27]	x		English
16	Yoon et al. (2015)	2.0%	1,500	199	199	43 [13 - 60]			English
17	Sander-Montant et al. (2023)	1.8%	1,381	123	123	22 [12 - 31]	x		English, French
18	Mahr et al. (2015)	1.6%	1,232	29	29	21 [18 - 24]	x		English
19	Egger et al. (2020)	1.4%	1,052	54	54	18 [18 - 19]	x		Dutch
20	Garrison et al. (2020)	1.4%	1,046	35	35	15 [12 - 18]	x		English
21	Hurtado et al. (2007)	1.3%	1,010	49	49	24 [15 - 37]			Spanish
22	Pomper & Saffran (2017)	1.0%	785	82	82	17 [14 - 19]			English
23	Frank et al. (2016)	1.0%	781	108	108	35 [12 - 60]			English
24	Bacon & Saffran (2022)	1.0%	754	38	38	23 [22 - 24]	x		English
25	Ronfard et al. (2022)	1.0%	737	40	40	20 [18 - 24]	x		English
26	Perry & Saffran (2017)	0.9%	718	42	42	20 [19 - 22]	x		English
27	Casillas et al. (2017)	0.9%	651	23	23	32 [9 - 48]			Tseltal
28	Swingley & Aslin (2002)	0.8%	596	50	50	15 [14 - 16]	x		English
29	Pomper & Saffran (2016)	0.6%	471	60	60	44 [41 - 47]			English
30	Kremin et al. (2023)	0.5%	410	60	60	42 [37 - 49]			English, French, Spanish
31	Pomper & Saffran (2015)	0.5%	389	25	25	40 [38 - 42]			English
32	Moore & Bergelson (2022)	0.5%	384	29	29	18 [16 - 20]	x		English
33	Potter et al. (2019)	0.4%	309	44	23	23 [18 - 29]	x	x	English
34	Byers-Heinlein et al. (2017)	0.4%	284	48	48	20 [19 - 21]	x		English, French
35	Pomper & Saffran (2019)	0.3%	249	44	44	40 [38 - 43]			English
36	Pomper & Saffran (2018)	0.3%	217	37	37	39 [38 - 43]			English
	Total	100%	75,636	4,851	3,171	22 [6–60]	21	5	

moment (indexed by LWL) and either concurrent vocabulary size or the growth of early vocabulary over time (indexed by the MB-CDI) (Frank et al., 2026).[2] Since some studies used different versions of the CDIs (e.g., short vs. long forms), all scores were converted to a proportion of the possible items.

All code and data to reproduce the analyses is available at https://github.com/peekbank/peekbank-method.

Data processing recommendations

Recommendation 1: Measure accuracy over a long analytic window.

Accuracy, or proportion of target looking, must be computed over a time window of interest. Typically, windows start shortly after the onset of the target word and continue until a few seconds later; this window is intended to capture the peak of children's stimulus-driven looks to the target. However, in practice, studies vary in what windows are

[2] We do not explore relations at the item-level (e.g., speed of processing of a particular word and a child's reported knowledge of that word) given the mixed evidence for item-level alignment between LWL and MB-CDI data (López Pérez, Moore, Sander-Montant, & Byers-Heinlein, 2024; Weaver & Saffran, 2026).

used, and often window selection is a locus for analytic flexibility (choosing windows post hoc to maximize a particular effect) (Peelle & Van Engen, 2021). Common analysis windows include 300–1800 ms after target word onset (following Fernald et al., 2008) or 367–2000 ms after target word onset (e.g., Swingley & Aslin, 2002), although windows that extend until 3 or 4 seconds after target word onset are also used, especially at younger ages (e.g., Bergelson & Swingley, 2012; Bergelson & Aslin, 2017; López Pérez et al., 2024). In this analysis, we begin from first principles and ask when windows should start and end to maximize reliability and validity on word recognition trials.

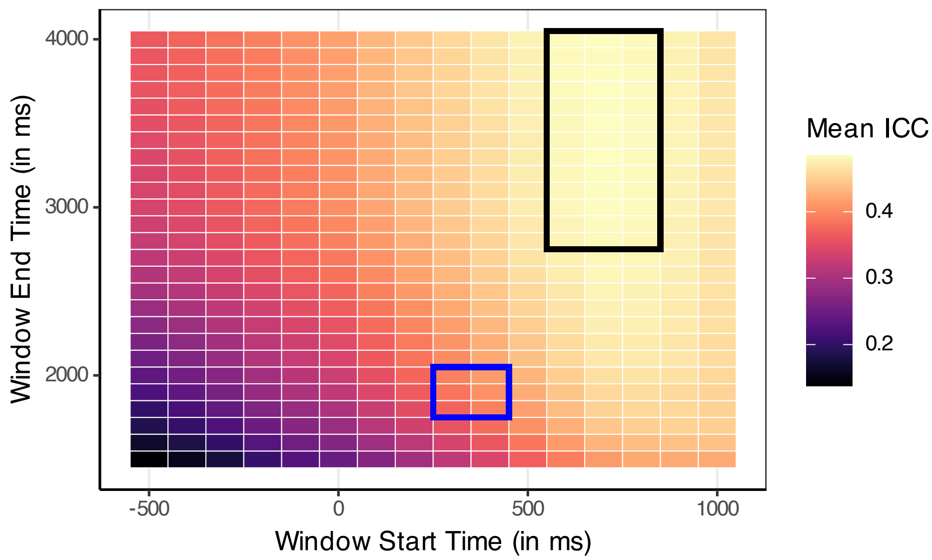

Figure 1. A heat map showing the average ICC across datasets for different start and end points for the accuracy window. A region with the highest ICC (best reliability) is outlined in black, and the region most commonly represented in the literature is outlined in blue.

Analytic approach. We first calculated the ICC reliability for accuracy across datasets, computing average ICC across a range of different analysis window start and end points. The standardized Peekbank format allowed us to apply each combination of window start and end points consistently within each dataset. Next, we added the additional metrics of test-retest reliability and – to assess validity – CDI correlations, which were calculated by dataset for each combination of parameter values. To aggregate across datasets, we used a linear mixed-effects model with random intercepts by dataset, weighted by the number of sessions. We interpret the fixed-effect contrasts as the predicted difference in outcome when switching from one set of parameters to another on a generic dataset. The best data-processing options might be different for children of different ages, so we ran the same linear mixed-effects models separately for children in each of 4 age groups: <18 months, 18–23 months, 24–35 months, and >=36 months.

Results.

Reliability. Average ICCs across datasets for different windowing options are shown in Figure 1. As expected, including time points from before the target word onset decreases reliability. Our initial exploration found a region of high reliability with start

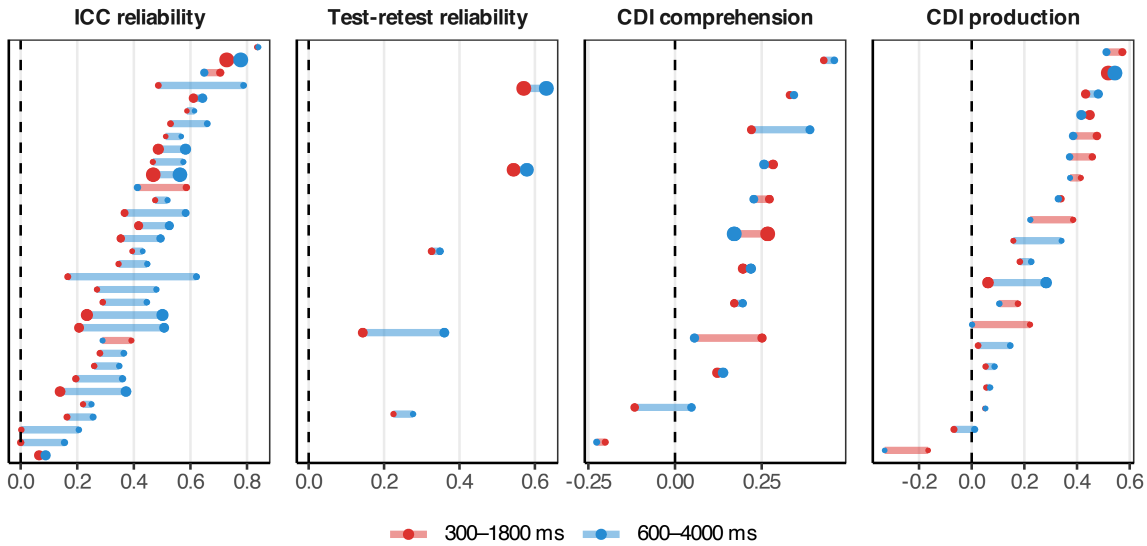

Figure 2. Per dataset outcome measures for two window options: 300–1800 ms (red) and 600–4000 ms (blue). Line color indicates whether the longer (blue) or shorter (red) window leads to a higher value. Dot size is proportional within each panel to the number of sessions.

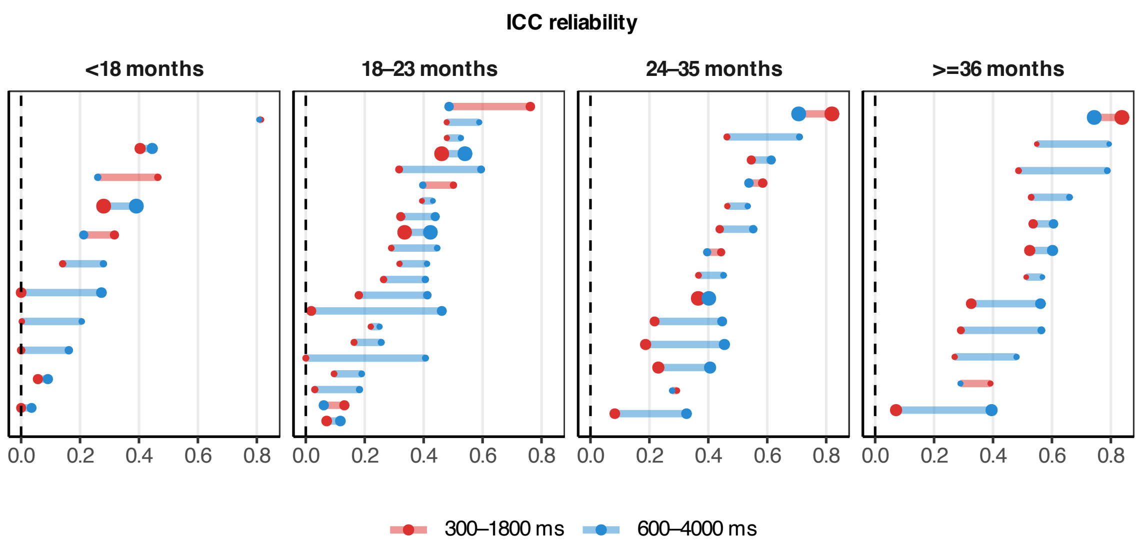

Figure 3. Per dataset ICC measures for two window options: 300–1800 ms (red) and 600–4000 ms (blue), split by age category. Line color indicates whether the longer (blue) or shorter (red) window leads to a higher value. Dot size is proportional within each panel to the number of sessions.

times around 600–800 ms and end times around 2800–4000 ms. A dataset by dataset comparison of the differences between a 300–1800 ms window (representative of common practice) and a 600–4000 ms window are shown in Figure 2. Nearly all datasets have higher ICC reliability for the later and longer window. All 5 test-retest reliability datasets also show increased reliability on this later and longer window. Our mixed-effects model confirms that there are expected reliability gains from switching from a 300–1800 ms window to a 600–4000 ms one. ICC reliability is expected to increase by 0.12 [0.08, 0.16] (square brackets indicate 95% CIs) and test-retest reliability by 0.07 [-0.02, 0.16].

These patterns generally hold across age. The overall level of reliability rises with age, as older children generally fixate target images more consistently and experiments with younger children typically use fewer trials (Figure S3). However, the overall pattern of which time windows maximize reliability at any given age is remarkably consistent across the different age groups (Figure S5). For instance, across different ages there is an improvement in ICC for using a 600–4000 ms window compared to a 300–1800 ms one (Figure 3). For <18 months, the ICC difference is 0.06 [-0.03, 0.16]; for 18–23 months, 0.13 [0.07, 0.19]; for 24–35 months, 0.06 [0.01, 0.12]; and for >=36 months 0.15 [0.06, 0.23].

Validity. There is no evidence of differences in validity between a 300–1800 ms window and a 600–4000 ms window (Figure 2, right 2 panels). Switching to the longer window is associated with a change of -0.02 [-0.07, 0.03] in correlation with CDI comprehension and 0.03 [-0.02, 0.08] with CDI production.

Discussion. Overall, we find that LWL accuracy measures are reliable across a wide range of window choices; however, to maximize reliability, we recommend windows that start somewhere in the 400–600 ms range and extend to 3000–4000 ms (longer than the canonical window that ends at 1800 or 2000 ms), regardless of child age. Windowing choices seem to primarily affect reliability measures and have limited effects on our validity measures. The finding that later starting and later ending windows increase reliability suggests that, even as looks to the target decline after 2000 ms, looking behavior still contains reliable signal. While some children may look quickly to the target and then look away, other children who look to the target later or who continue looking at the target are providing a useful signal during later portions of a given trial. Overall, children in many datasets continue to show consistent looking to the target over the distractor during later trial periods (Figure S2). An implication of these results is that choosing analysis windows based on the peak of the grand average proportion target looking time course generally leads to lower reliability estimates, with no gains in validity.

Recommendation 2: Do not use baseline-corrected accuracy.

One commonly acknowledged issue in LWL studies is that looking behavior could be driven by differences in visual salience between the target and distractor images themselves (Fernald et al., 2008; Novack et al., 2025): Children may have a baseline preference to look at one image over another, even in the absence of auditory stimuli. Although thoughtful design choices can minimize these issues (see Recommendation 6), some analytic strategies also attempt to account for baseline preferences in dependent measures. While there are a variety of such strategies in the LWL literature, here we focus on one common practice: baseline correction. In baseline correction, researchers use a difference measure between accuracy on a critical post-stimulus window and “accuracy” (proportion looks to target among looks to target or distractor) on a pre-stimulus baseline window (e.g., Bergelson & Swingley, 2012; Weaver, Zettersten, & Saffran, 2024). The baseline window can be the 2 seconds prior to target word onset (Weaver et al., 2024) or the full 3–4 second pre-stimulus period (Bergelson & Swingley, 2012).

Analytic approach. We considered baseline-corrected accuracies with various pre-stimulus baseline windows, focusing on a 600–4000 ms critical window (although the results generalize to other critical windows). We restricted analysis to datasets with at least some data from 2000 ms or more before target word onset. We also computed a "no baseline" accuracy, based solely on accuracy within a 600–4000 ms time window with no correction. Baseline windows started at -1000, -2000, -3000 and -4000 ms and ended at target word onset.

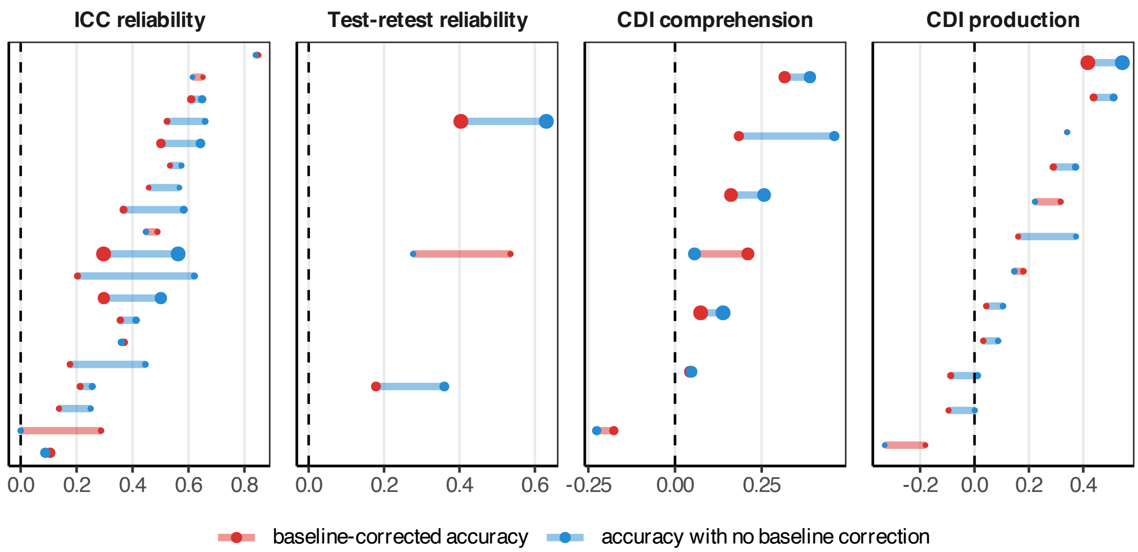

Figure 4. Comparison between a baseline-corrected accuracy in red and accuracy in blue across datasets and measures. Both use a critical window of 600–4000 ms; the baseline window was -2000–0 ms. Line color indicates whether the longer (blue) or shorter (red) window leads to a higher value. Dot size is proportional within each panel to the number of sessions.

Results.

Reliability. Baseline-corrected accuracy led to worse reliability than critical-window accuracy alone. As an illustrative example, Figure 4 shows per-dataset results for accuracy compared to a baseline-corrected accuracy using a -2000–0 ms baseline window. This result held generally: All baseline window sizes that we considered led to worse reliability compared to using normal accuracy without any correction. For a critical window of 600–4000 ms, baseline correction is expected to decrease ICC by at least 0.10 [0.04, 0.16] and test-retest reliability by at least 0.19 [0.08, 0.30]. Similar patterns hold for different accuracy windows (Figure S6) and across different age groups (Figure S8).

Validity. Baseline-correction also reduces the validity of accuracy measures (Figure 4, right panels). All baseline window sizes that we considered led to worse correlations with productive vocabulary (though results were less robust for receptive vocabulary). For a critical window of 600–4000 ms, baseline correction is expected to decrease the correlation with receptive vocabulary by at least 0.04 [-0.05, 0.12] and correlation with productive vocabulary by at least 0.09 [0.05, 0.13]. Similar patterns hold for different accuracy windows (Figure S6) and across different age groups (Figure S8).

Discussion. We do not recommend the use of baseline-correction in accuracy measures, as it worsens both reliability and validity compared to a critical-window-only accuracy measure. Why would incorporating baseline information lead to worse metrics? The literature on the psychometrics of difference scores (e.g., Lord, 1956; Rogosa & Willett, 1983; Trafimow, 2015) provides a valuable guide for evaluating the expected measurement properties of baseline correction. This literature points to two important features of LWL tasks: first, children's looking behavior is noisy from trial to trial, meaning that the reliability of proportion looking during both the baseline and the critical window is low; and second, the signal across baseline and critical portions of a trial is typically correlated (in part due to the very salience that the baseline correction is trying to remove). Unfortunately, it is under precisely these conditions – low reliability and high correlation between components – that difference scores exhibit the lowest reliability (Trafimow, 2015). Thus, while baseline-correction may lead to grand average accuracy scores that are less biased (by removing item salience), it also leads to less reliable and valid individual-level measures.

Recommendation 3: Measure landing reaction times with a minimum reaction time of around 400 ms.

Reaction time (RT) reflects the speed with which infants shift their gaze from the distractor to the target image. RT computations include many choice points. RTs can be computed based on when a child's eyes leave from the distractor (launch RT) or when a child's eyes arrive on the target image (landing RT) (Fernald et al., 2008). A minimum and maximum RT may be set, as short and long RTs are often excluded because they are considered unlikely to be generated in response to the stimulus (Fernald et al., 2008). Reaction time data are sometimes analyzed as raw scores and sometimes analyzed using a logarithmic transform (consistent with the idea that reaction times are log-normally distributed and that effects on them may be multiplicative rather than additive) (Csibra, Hernik, Mascaro, Tatone, & Lengyel, 2016; Frank et al., 2026). We systematically investigated the effect of each of these choice points on reliability and validity.

Analytic approach. We considered a wide range of options for measuring RT, fulling crossing a) launch or landing RTs, b) a minimum RT threshold for inclusion between 0 and 1000 ms, c) a maximum RT threshold for inclusion between 1600 and 4000 ms, and d) raw or log-transformed RTs.

Results.

Reliability. An initial exploration indicated that ICC reliability was maximized for landing RTs with a minimum RT around 400 ms and a high maximum RT (Figure 5). A comparison between a common set of parameters for calculating RT and an alternative that maximizes ICC are shown in Figure 6. Compared to computing the "standard" raw launch RTs restricted to 300–1800 ms, using log-scale landing RTs ranging between 400–4000 ms is expected to increase ICC by 0.11 [-0.01, 0.23], with little change in test-retest reliability (0.04 [-0.18, 0.27]). Similar patterns hold for children of different ages (Figure S9 and Figure S11).

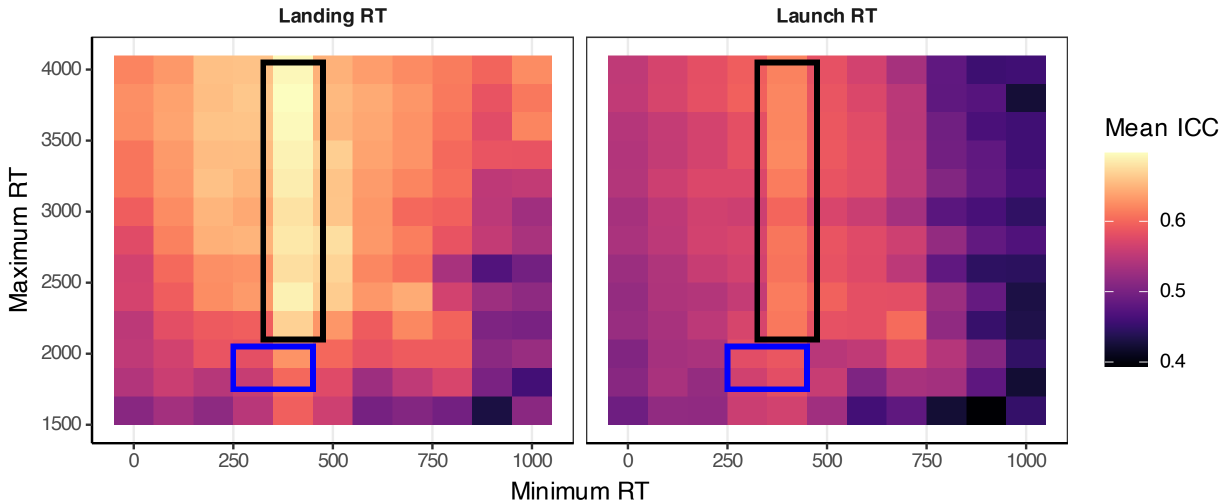

Figure 5. Heat map of different parameter options for measuring RT and how they on average affected ICC. A region with the highest ICC (best reliability) is outlined in black, and the region most commonly represented in the literature is outlined in blue.

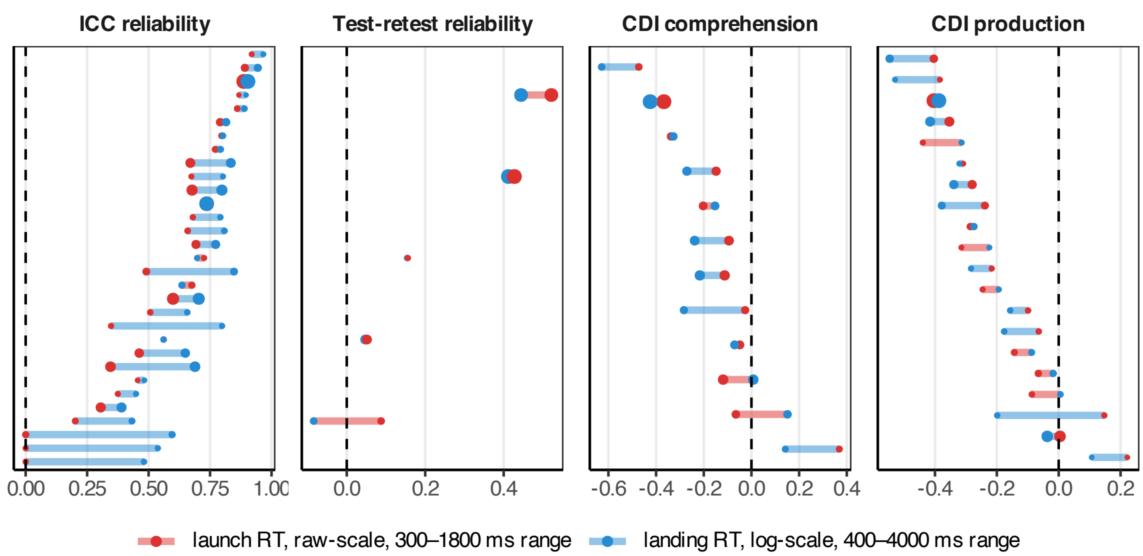

Figure 6. Comparison between two different sets of RT parameters. In red, an example of common practice, using a launch RT from 300–1800 ms. In blue, an RT that maximizes ICC, with a landing RT from 400–4000 ms, on the log scale. Note that correlations with the CDI are generally expected to be negative (as faster word recognition is associated with larger vocabulary). Line color indicates whether the longer (blue) or shorter (red) window leads to improved reliability or validity.

Validity. Different methods for calculating RT had little impact on validity (Figure 6, right panels). Compared to computing the "standard" launch, raw RTs restricted to 300–1800 ms, using landing log RTs ranging between 400–4000 ms is expected to have little impact on CDI correlations (-0.04 [-0.17, 0.08] for comprehension, and -0.03 [-0.12, 0.06] for production).

Discussion. Overall, RT measures capture substantial signal, as indicated by the moderate ICC values across all choice parameters. A wide range of parameter options have similar properties (Figure S10 and Figure S12), but computing landing RTs with an inclusion window of 400–4000 ms maximizes reliability without negatively impacting validity. Note that the recommendation to compute landing reaction times contradicts traditional approaches to reaction-time computation (Fernald et al., 2008) based on launch time; however, landing RTs generally lead to higher ICCs in our dataset. Log-scale transformations should be considered, but may depend on modelling and theoretical considerations.

Recommendation 4: Trial-level and session-level exclusions for low amounts of data are generally unnecessary.

A common issue in developmental research is when to apply trial- and session-level exclusions. For LWL studies, trial-level exclusions tend to be applied when children contribute too few time points (e.g., due to tracker loss or looking away from the screen during the trial). Papers often use a combination of heuristic exclusions for inattention prior to coding looking behavior (Adams et al., 2018) and criterion-based exclusions based on the proportion of looking to the target and distractor during the trial. Cutoffs vary, with different experiments requiring minimum on-screen looking for at least 20%, 50%, or 66% of the critical window of a trial (Borovsky & Peters, 2019; Pomper & Saffran, 2019; Weaver et al., 2024). Studies also occasionally filter for on-screen looks specifically at the point of disambiguation (Potter, Fourakis, Morin-Lessard, Byers-Heinlein, & Lew-Williams, 2019).

In addition, studies also implement session-level exclusions based on the minimum number of valid trials. For accuracy measures, children are often excluded if they had fewer than 25% or 50% of the total trials in a session, either within a condition, or overall (Adams et al., 2018; Garrison, Baudet, Breitfeld, Aberman, & Bergelson, 2020; Perry & Saffran, 2017). For RT, it is instead standard to require a minimum number of trials (e.g., two, Fernald et al., 2006; Adams et al., 2018).

There are multiple possible ways that trial- and session-level exclusions could affect measurement properties. If trials with fewer valid looking points and sessions with only a small number of trials still provide signal of the child-level language behavior, excluding them will decrease reliability and validity. On the other hand, if these trials and sessions reflect noise, excluding them will increase reliability and validity.

Analytic approach. We consider 3 types of exclusions: trial-level exclusions for accuracy, session-level exclusions for accuracy, and session-level exclusions for RT. We first explored trial-level accuracy exclusions, both on the basis of proportion of off-screen looks and on the basis of where the child was looking at target word onset. For accuracy, we then considered per-session minimums either based on the number of valid trials, or based on the fraction of valid trials in a session. The fraction of valid trials is the more common exclusion in the literature, but when drawing from many datasets that vary in the total number of trials, minimum trial count likely represents a more comparable metric. For example, a 50% exclusion criterion implies a different number of valid trials for an experiment with 16 vs. 32 total trials (a minimum of 8 vs. 16 valid trials, respectively); criteria based on valid trial counts therefore result in a more consistent approach to exclusions across datasets. For RT, we considered different cutoffs for the minimum number of valid trials required for inclusion.

We note that datasets contributed to Peekbank vary in whether excluded data was contributed, or whether only cleaned post-exclusion data was contributed. Thus, some datasets may not show effects of exclusions if those trials or sessions that would be excluded are missing from Peekbank.

Results.

Reliability. For trial-level exclusions, a requirement to look on-screen at least 50% of the time results in 85.64% (range: 27.22%–98.92%) of trials being retained. These exclusions lead to an expected increase in ICC by 0.03 [0.00, 0.05] and no substantial change in test-retest reliability (-0.02 [-0.11, 0.08]). Filtering based on initial looking position, or having requirements for looking stricter than 50%, start to diminish the amount of data and potentially negatively affect test-retest reliability (Figure S13 and Figure S14).

We then explored the effects of excluding sessions for too few trials. Here, we focus on the results for a minimum number of 8 total trials, with no trial-level exclusions, to illustrate general patterns in the results, but report outcomes for a range of proportion- and count-based thresholds in the supplement (Figures S15–S18). Only including sessions with at least 8 trials results in retaining 80.99% (range: 0%–100%) of total sessions. These exclusions show little evidence of increasing ICC (0.00 [-0.01, 0.02]) and or test-retest reliability (0.03 [-0.08, 0.14]) compared to retaining all sessions.

We also considered the effects of minimum trial requirements on RT data. Only including sessions that contribute at least 2 trials results in retaining 90.93% (range: 42.86%–100%) of all sessions. Similar to previous results, these exclusions lead to small changes in reliability (ICC: -0.01 [-0.07, 0.04]; test-retest reliability: 0.03 [-0.08, 0.14]).

Validity. Excluding trials with less than 50% on-screen looking did not substantially change the validity of accuracy measures compared to no exclusions (comprehension: 0.01 [-0.04, 0.06]; production: 0.00 [-0.03, 0.03]). As with reliability, on-screen looking criteria stricter than 50% increase data loss and consequently may even have negative effects on validity (Figure S13 and Figure S14). Similarly, session-level exclusions did not substantially impact validity. Only including sessions with at least 8 trials, compared to including all sessions, led to virtually no increase in correlations (comprehension: 0.01 [-0.04, 0.07]; production: 0.01 [-0.03, 0.05]). For RT, excluding sessions with fewer than 2 valid RT trials also led to no substantive changes in correlations with CDI (comprehension: 0.00 [-0.15, 0.14]; production: -0.02 [-0.08, 0.04]).

Discussion. Overall, exclusions based on amount of data per-trial or per-session come with trade-offs between the amount of data and variability within the data. Standard practice within the field often errs on the side of excluding a large number of trials and sessions for missing data. Surprisingly, our results show that applying no exclusions or minimal exclusions often represents a reasonable choice, at least in terms of the psychometric properties of the resulting measurements.

Design recommendations

Our focus for this work was to explore how data processing choices affected the reliability and validity of the resultant measures. However, along the way, we identified several experimental design questions that had notable consequences for our central measures. We summarize two design recommendations that stood out as having the largest potential impact on reliability and validity (see supplement, section S7 for an additional analysis of order effects).

Recommendation 5: Design studies to have a minimum of 10 trials per child, per condition, or as many trials as possible for individual-difference studies.

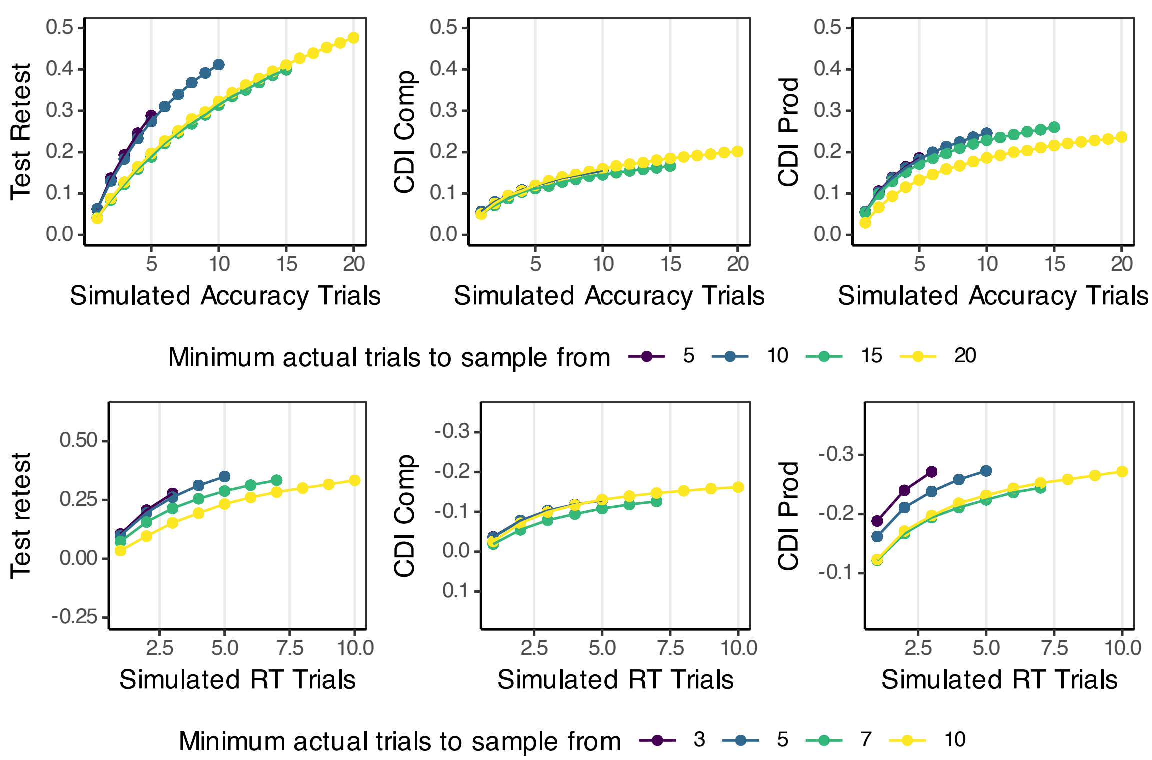

Figure 7. Simulations of measurement properties for different trial counts, created by down-sampling trials for all children with different trial minimums. Shown for an accuracy window of 600–4000 ms with no trial- or administration- level exclusions.

We found limited evidence for excluding children for low trial counts, but intuitively, more repeated measures should improve measurement properties such as reliability and validity. Rather than retrospective exclusions, another approach is prospective: how many trials per child should one aim for to achieve adequate reliability and validity? Test-retest reliability is important for studies that investigate longitudinal effects, as it assumes that the test sessions are representative of properties of the child at that age (Byers-Heinlein et al., 2022). To address this prospective question, we considered how reliability and validity measures would have changed if we had fewer trials per child. We filtered the data to include only children who had contributed at least N trials (for different values of N) and then repeatedly subsampled their trials, calculating metrics for each subset. Datasets vary in how many trials they have, so different minimum trials include different distributions and datasets, items, and children.

The results for accuracy and RT are shown in Figure 7. ICC reliability is not sensitive to the number of trials per subject, so it is not shown. For accuracy, more trials per session increases test-retest reliability, up to 20 trials (the highest we measured), although the gains per additional trial decrease with more trials. CDI-based validity also improved as trials increased, with smaller gains past 5 trials. Based on accuracy, we recommend at least 10 trials per condition in general, and as many as possible for individual differences studies where test-retest reliability is essential.

Similar considerations apply for RT, but at lower total trial numbers. Nearly every trial a child does results in a usable accuracy measure, but only around 1/3 of trials result in a valid RT measure in our data. For test-retest reliability, having at least 5 RT measures per session leads to notable improvements; gains beyond 5 are hard to measure because of the low number of subjects with sufficient trials, but more trials appears to be beneficial. For CDI measures, there are substantial gains for having 2 trials compared to 1 and continued benefits to including additional trials. Overall, for measuring individual differences in processing speed, designing studies to include as many trials as possible appears to be central to adequate measurement.

Recommendation 6: Match target and distractor images based on animacy.

One concern with LWL trial design is that children may have preferences for which image to look at separate from the auditory stimuli, and some of their looking behavior may be independent of the auditory stimulus. One approach is to decrease the preference discrepancy between a pair of images, so that in the absence of auditory cues, children prefer to look at both images approximately equally. While there are a number of factors that could play into children's image preferences (Fernald et al., 2008; Nothdurft, 2000; Pomper & Saffran, 2019), one factor that stood out in the data was animacy. Children have a strong preference for looking at animals and humans. When trials have one animate and one inanimate, children look more at the animate in the pre-stimulus period (Figure 8A) and are slower to respond when the inanimate is named. In contrast, for trials with target and distractor items matched on animacy, children's early looking patterns for animate and inanimate targets are highly similar. This difference is particularly relevant for the younger children, whose stimulus-contingent looking behavior is weakest, so more of the looking behavior is driven by animacy; however, effects of animacy were apparent in children up to the age of three (Figure 8B).

This finding has important consequences for estimating word recognition: accuracy for animate targets is substantially overestimated and accuracy for inanimate targets becomes substantially underestimated when animacy is not balanced in the target-distractor pairing. The results are especially dramatic for younger infants (under 18 months): looking to familiar inanimate targets can appear to be at chance when paired with an animate distractor. In other words, an improper balancing of saliency could make an infant appear to not know a word, even though they do.

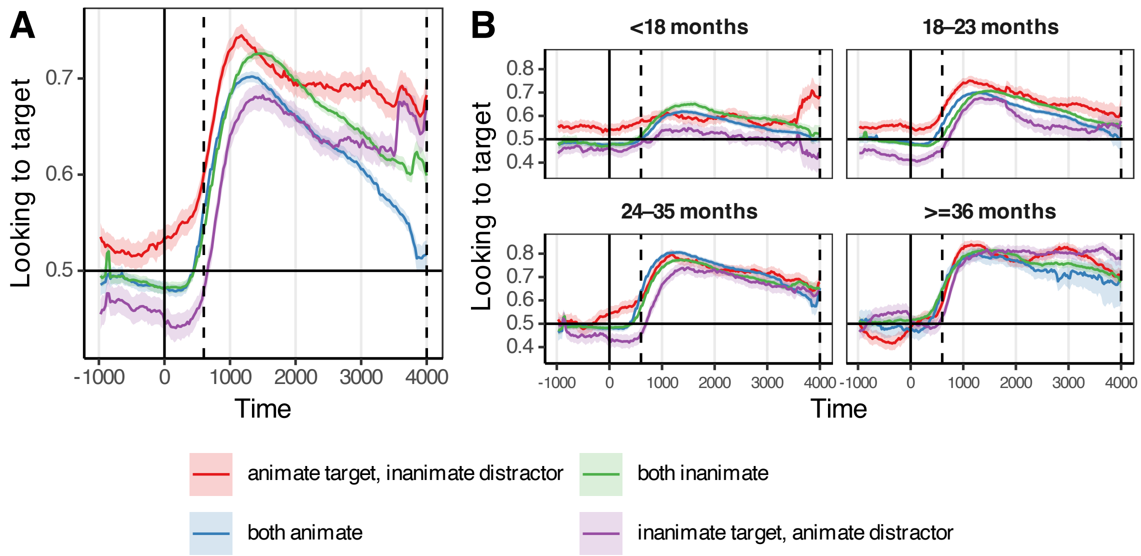

Figure 8. Looking time profiles for different combinations of target and distractor animacy. Panel A shows overall; Panel B shows split by age category.

How much can this matter?

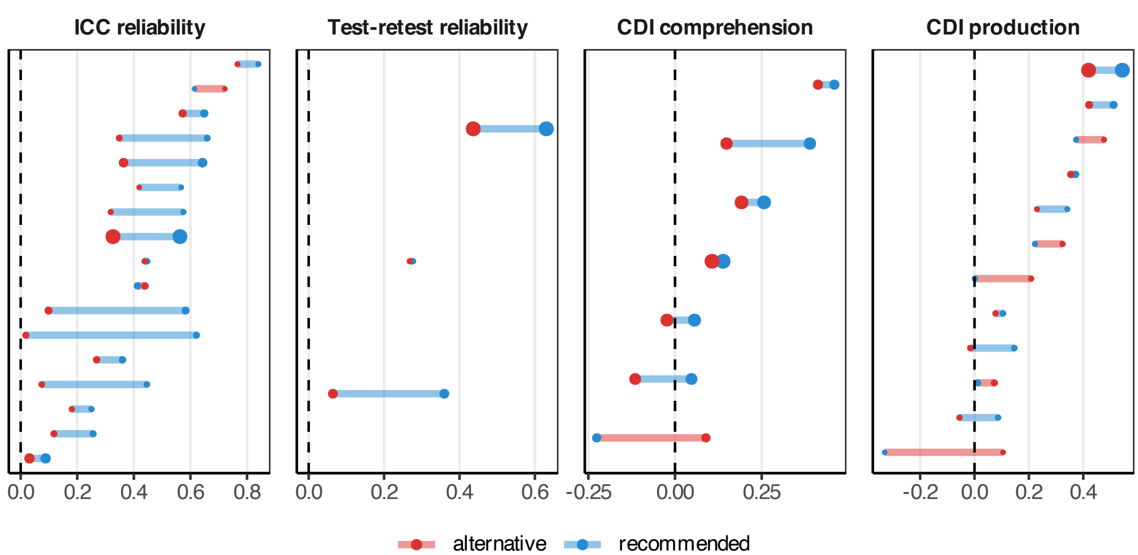

Figure 9. Differences in dependent measures between the recommended accuracy measurement and a sensible-sounding, but unfortunate alternative set of parameters.

To quantify the potential cumulative consequences of different combinations of choices, we computed the difference in reliability when making all of the recommended choices with respect to data processing to an instance in which a researcher made a set of unfortunate choices that are nevertheless consistent with the literature. Consider two researchers analyzing accuracy in the same LWL dataset, Norm and Rex. Researcher Norm makes a set of reasonable choices based on practices observed in other LWL papers, choosing to analyze a short analysis time window (300–1800 ms); apply baseline correction (baseline window: -2000–0 ms); exclude trials on which infants are not looking at the screen for at least 60% of the critical window or not looking at the screen during target word onset for insufficient attentiveness; and require that infants contribute at least 50% of all trials for inclusion. Researcher Rex, on the other hand, analyzes the same data and instead chooses parameters based on the data-driven recommendations above: a critical window of 600–4000 ms, no baseline correction, and including all trials and sessions. Researcher Rex would gain substantial increases in ICC (0.20 [0.13, 0.27]) and test-retest reliability (0.21 [0.13, 0.29]). Additionally, Rex could expect small increases in validity for correlations with comprehension (0.07 [-0.03, 0.17]) and production (0.07 [0.00, 0.13]). Despite both researchers making choices consistent with past practices, we would expect Rex to derive a measure with improved psychometric properties compared to Norm, and therefore be in a better position to study individual differences and longitudinal change.

General Discussion

Children's early word recognition is a central construct for the study of language learning, so improving recognition measures can have widespread benefits for our understanding of language development. How can we maximize reliable, valid signal in the analysis of data in the LWL paradigm? We took a data-driven approach to this question, using a large-scale dataset of infant LWL data to quantify the impact of different analysis choices on measurement properties, leading to six core recommendations for data processing and experimental design.

For data processing, our strongest take-homes concern window choice and baseline correction. First, longer time windows (e.g., windows that end around 3000 ms after target word onset) maximize the reliability of accuracy measures, with no negative consequences for validity. While average accuracy tends to peak earlier in the timecourse of a trial (typically 1000–2000 ms after target word onset), these findings suggest that later looking windows capture signal that boosts the consistency of word recognition measures. Second, baseline correction, while an intuitive solution to imbalances in the saliency of images, causes large decrements in reliability and moderate decrements to validity, and we therefore recommend avoiding this practice. Finally, we formulate recommendations for choosing reaction time cutoffs (around 400 ms tends to maximize reliability) and find that implementing trial- and session-level exclusions based on the amount of data contributed tend to have minimal consequences. Overall, choices about windows of analysis, approaches to exclusions, and how scores are computed can all have meaningful impacts on measure reliability.

One question that emerges from our analysis concerns the nature of the construct that LWL measures. In our analysis, long accuracy windows – as well as our other recommendations – increased reliability more than validity. Without more detailed characterizations of individual children, we cannot offer definitive answers to this question, but there are several possible, non-mutually exclusive, explanations. One possibility is simply that some aspects of this difference relates to measurement error in CDI scores, which attenuates validity. Second, although the CDI is one useful source of validity evidence, it is possible that there are components of in-the-moment language processing that are not related to the primary construct captured by the CDI, which is general word knowledge. Finally, perhaps some aspects of our recommendations boost reliability in part because they capture non-linguistic variation between children, for example differences in attention or processing speed. For example, perhaps signal from later in the accuracy timecourse (which boosts reliability) might be less coupled to language per se, and might instead reflect more general features of attention. Future research could use this difference as a starting point to further parcellate the attentional and linguistic contributors to language-driven eye-movements.

A more general take-home from our data processing analyses is that looking at patterns in grand-average time courses is not a trustworthy guide to child- or item-level psychometric properties. Past recommendations for time window selection have sometimes been motivated by inspecting consistent patterns of looking in grand-averages and noting when looking begins to decline across participants. These heuristics – while based on important patterns in looking behavior – may not always lead to optimal choices with respect to core measurement properties such as reliability. Wherever possible, measurement choices should be guided by direct estimates of psychometric properties in order to maximize the signal in infant looking behavior (Byers-Heinlein et al., 2022; Sander-Montant et al., 2026; Weaver et al., 2026).

Among our design-focused recommendations, the most striking result is the impact of animacy on infant looking behavior. When an image of an animate referent was paired with an image of an inanimate referent, infant looking was systematically biased towards the animate image, even after target word onset. These results emphasize the importance of balancing saliency in image pairings as much as possible in LWL designs, particularly given given that many of the current analytic approaches to correcting for saliency imbalances post hoc rely on difference-based measures which decrease reliability.

The second design recommendation is to include as many trials as possible when estimating effects for individual children. Our recommendations of at least 10 trials per condition for accuracy and at least 5 trials for reaction-time are based on the substantial gains in reliability and validity achieved by meeting these minimum thresholds. However, reliability and validity continue to increase monotonically as the number of trials per participant increases; for estimating individual-level effects, maximizing the number of trials collected per infant is therefore critical. These findings converge with other recent results in the developmental literature emphasizing the importance of increasing per-participant observations in individual differences research (Byers-Heinlein et al., 2022; DeBolt, Rhemtulla, & Oakes, 2020; Schreiner et al., 2024). Increasing trial numbers represents a significant challenge for developmental research, given the limits infant attention place on the duration of single test sessions. One potential solution is to distribute trials across multiple testing sessions per infant (Adams et al., 2018; Weaver et al., 2024); this approach has proven successful, but is also time- and resource-intensive. Regardless of the approach, choosing study designs that maximize trial numbers per participant will remain central to estimating individual differences and studying longitudinal change (Weaver et al., 2026).

Limitations & Generalizability

The scope of our recommendations is fundamentally limited by the characteristics of the Peekbank dataset. First, while the Peekbank database is by far the largest dataset of its kind, it is not a comprehensive or representative set of LWL studies. In particular, (1) the data is predominantly English word recognition collected from infants living in the U.S., and (2) the dataset is a convenience sample from labs who made their data openly available or otherwise accessible on request from the Peekbank team. LWL studies as a whole are likely more heterogeneous than what is represented in Peekbank – the datasets considered here come predominantly from a small network of researchers with similar training in LWL data collection.

At the same time, there is also substantial heterogeneity in the studies included in Peekbank, as different datasets were designed with different research questions in mind. To mitigate this concern, our analyses focused exclusively on standard trials that present familiar words with familiar images with limited or no targeted manipulations of the stimuli. The main benefit of focusing on standard trials is that it allows for a clear interpretation of reliability and validity – trials are truly designed to measure word recognition, rather than some other aspect of language processing. We view the remaining heterogeneity of study properties (e.g., specific words; differences in age) as a strength of our approach – our findings generalize broadly across a wide range of items and ages. However, by focusing on standard trials only, we also limit our conclusions to measuring familiar word recognition; our findings do not address how data processing and design practices would generalize to other estimation goals, e.g., estimating the magnitude of differences between two experimental conditions. However, some of the more consequential choices, e.g. baseline correction, are likely have a similar impact given general analysis principles known to affect reliability (e.g., computing difference scores). In general, our recommendations can be seen as a reasonable default choice when analyzing looking-while-listening data in the absence of data-driven guidance tuned to one's specific estimation goal.

While we summarize the choices that typically maximize reliability, we note that we often see a range of choices with similar outcomes in our analyses. There is no one choice of time window that we can definitively infer as the "best" for maximizing reliability – instead, we see a pattern of graded improvements in reliability that follow a predictable pattern. For example, longer time windows tend to increase reliability holding the start of the accuracy window constant. While we sometimes articulate which choices lead to the highest reliability descriptively, these are at best approximations that we offer as data-driven default choices (especially in cases in which the decision would otherwise be ad hoc).

Conclusions

Experimental design and data processing decisions can have profound impacts on the properties of infant measures. Our findings show how compiling large-scale datasets such as Peekbank can help us estimate the impact of a wide variety of experimenter decisions, generalizable across a wide range of ages, stimuli, and experimental contexts. Establishing consistent practices and principles for reliable and valid measurement of infant looking sets the stage for asking central questions about how language skill develops.

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